F24-173-D-CyberQuest

FYP Team

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Session 2021-2025

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October, 2024

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Chapter 1

Introduction

Teens and young adults in today's digital environment rely extensively on the internet for nearly every aspect of their everyday life. The internet has become an essential component of people's social, educational, and personal lives, whether they are using it for gaming, shopping, or social media connections with friends.

Common online activities include:

- **Social Connectivity:** For young people to remain in touch, share their lives, and express themselves, social media, messaging applications, and online communities are essential.
- **Educational Resources:** Digital tools and resources are now crucial for academic performance, from collaborative projects to online courses and research.
- **Entertainment and Gaming:** A lot of teenagers and young adults use streaming services, online games, and interactive entertainment to pass the time when they have spare time.

The demand for effective cybersecurity measures rises in parallel with the growth of internet usage. Cybersecurity is defending networks, systems, and data from intrusions and attacks by unauthorised parties. Ensuring the security of personal information, maintaining privacy, and safeguarding users against persistent dangers in the digital realm are vital. **?**.

1.1 Problem Statement

The problem of people not being aware of potential vulnerabilities could lead to cyber-attacks. Teenagers are especially susceptible to attacks because they lack the awareness needed to identify and counter cyber threats. Teenagers and young adults are more susceptible to cybercrimes, such as ransomware attacks and social engineering attacks, in underdeveloped areas, such as South Asia, which encompasses the Indo-Pak region.

Financial loss, identity theft, and other negative outcomes are possible outcomes of these attacks. As a result, people could participate in unsafe online activities or become the targets of cyberattacks, putting others and institutional data at risk.

Existing methods for spreading cybersecurity awareness, including traditional educational programs and awareness campaigns, often fail to engage audiences effectively. These approaches tend to be uninteresting and lack the interactivity needed to capture attention, resulting in a lack of motivation among people to learn and apply essential cybersecurity practices. As a result, many individuals are reluctant to engage with these resources, leaving significant gaps in their understanding and preparedness against cyber threats.

1.2 Scope

Project Focus: A game-based intervention for cybersecurity awareness using Unity 3D and virtual reality.

Target Audience: Teenagers and young adults.

Objective: Design, develop, and implement a 3D game to increase knowledge of cybersecurity threats and best practices in order to increase knowledge and awareness of cybersecurity threats and best practices in a realistic and engaging setting.

Learning Environment: To make learning relatable and interesting, the game will include a virtual environment that mimics real-world locations including homes, restaurants, and institutions. It will include virtual labs, classrooms, and scenarios that demonstrate these attacks and teach preventive measures, while also providing solutions after falling prey to a cyber attack. The scope consists of both the technical and educational aspects of the game.

Educational Scenarios: Demonstrates various cyber attacks through interactive scenarios. Teaches preventive measures against cyber threats. Provides solutions for players after experiencing cyber attacks.

Social Life Storyline: The game features a realistic storyline that captures the player's daily activities. Activities include getting on a bus, attending university, visiting restaurants, and other daily routines.

- **Player Role and Interaction:** The player enters the game as a character with established relationships with various non-player characters (NPCs) based on the story. Interactions occur with various non-player characters (NPCs) based on the storyline.

- **Reputation System:** The player develops a positive reputation through interactions with NPCs. Positive social interactions boost the player's reputation in front of others.
- **Rivals and Attackers:** The player faces rivals, such as fellow students, who seek to embarrass them. Multiple attackers appear as NPCs in different scenarios, adding challenges to the game.
- **Mood Bar System:** The game features a 'mood' bar with two moods: 'happy' and 'dead inside'. Positive social interactions increase the 'happy' mood, earning the player coins. Negative social points shift the mood towards 'dead inside'; if fully 'dead inside', the player loses that level.
- **Reputation Impact:** The player's reputation is influenced by their good or bad decisions. Good decisions enhance reputation and social points, while bad decisions and cyber attacks lead to negative outcomes, impacting how NPCs perceive the player such as leaked personal information.
- **Gameplay Progression:** The player navigates daily life scenarios as a teenager or young adult. Scenarios include attending university, visiting restaurants, and other daily activities. The game spans through all levels of 'University life' and concludes on the first day of the player's office life.
- **VR Integration:** The game uses VR technology to simulate real-life scenarios. Features include decision-making that affects scenario selection, virtual social interactions, and real-time feedback. Designed for deployment on various VR platforms, including Oculus headsets.
- **Cyber Attack Taxonomy:** The game includes common cyber attacks within its system boundary such as Phishing, Man-in-the-Middle (MitM) attacks, Brute force attacks, Malware, Ransomware, Email spoofing, USB-based attacks, Sniffing, Default password exploitation, Spam attacks, Social engineering. These attacks are implemented in daily life scenarios to educate players on their nature, prevention, and solutions.
- **Out-of-Scope Attacks:** Certain attacks are not covered as they fall outside the core objectives of the project such as DDoS, Clickjacking, Hash injection, ARP poisoning,DNS attacks, SQL injection. These attacks are considered irrelevant to the project scope residing in the irrelevant domain of the project.
- **Platform Support:** The game is optimized for VR platforms, specifically including Oculus VR headsets.
- **Graphics:** Graphics include realistic and immersive 3D models which accurately reflect daily life environments.

- **User Interface:** Includes simple and intuitive design, feedback mechanisms, easy navigation and clear suggestions for player assistance.
- **Audio:** Includes ambient sounds that reflect the game environment and effects that highlight key events and decisions.

Key functionalities within the project scope include:

- **Scenario Simulation**: It is the process of modeling actual cyberattacks, including ransomware, social engineering, and phishing.
- **Educational Content:** Interactive tutorials and feedback systems to reinforce learning.
- **VR Integration:** Support for various VR platforms, including Oculus Quest 2, to provide an immersive experience.

Localized Content: Scenarios and narratives tailored to the cultural context of the Indo-Pak region.

1.3 Modules

The game consists of several key modules, each focusing on different aspects of gameplay, from simulating cyber attacks to integrating VR technology. Below is the detailed description of each module, highlighting its main features.

1.3.1 Module 1: Game Mechanics

This module focuses on the core gameplay mechanics that define how players interact with the game environment. It includes the basic movement, actions, and interactions available to the player, ensuring a responsive and engaging gameplay experience.

1. **Feature 1: Player Movement and Controls**
   (a) Develop intuitive player controls for movement, such as walking, running, working, and jumping, optimized for both standard and VR gameplay.
   (b) Implement player actions like interacting with objects, allowing players to actively engage with scenarios and complete tasks.
2. **Feature 2: Interaction with Game Objects**

(a) Provide mechanisms for interacting with objects within the game and non-player characters (NPCs) to assist with task fulfillment and scenario-specific event triggering.

(b) Incorporate gameplay elements that are directly related to the game's cybersecurity themes, such as picking up things and hacking terminals.

1.3.2 Module 2: Game Elements

This module covers the game's visual and aesthetic components, such as the construction of the environments, characters, and objects. It seeks to define the game's aesthetic, making it aesthetically pleasing and thematically coherent.

1. **Feature 1: Visual Design of Environments**

   (a) Build realistic 3D models of buildings, interiors, and other structures to produce immersive experiences that mimic real-world locations like residences, public spaces, and university campuses.

   (b) Focus on realistic textures, lighting, and environmental effects to enhance the game's overall aesthetic and make the settings feel authentic and engaging.

2. **Feature 2: Aesthetics and Game Objects**

   (a) Design game objects, such as furniture, personal items, and digital devices, with a realistic and culturally relevant style.

   (b) Ensure that these elements contribute to the game's visual storytelling, supporting the narrative and the player's immersion in each scenario.

1.3.3 Module 3: Assets Creation (Environment and Characters

The visual assets of the game, such as the environments, props, and characters, are designed and created in this module. Its main goal is to produce an immersive visual experience that closely resembles actual environments.

1. **Feature 1: Development of characters**

   (a) This involves creating characters, such as rivals, allies, and friendly NPCs, with unique traits, lighting, and visual effects that correspond to their roles in the game.

2. **Feature 2: Development of Immersive Environments**

(a) Creating intricate settings, such as public areas, homes, and campuses of universities, with accurate lighting and textures to improve immersion.

1.3.4 Module 4: Game Scenarios

This module involves the development of storylines, levels, and narrative elements that guide the player through the game. It includes the design of scenario-based interactions and educational narratives that teach cybersecurity concepts through engaging gameplay.

1. **Feature 1: Storyboards and Narrative Building**

   (a) Make thorough storyboards for every scenario that describe the story's development and the player's travel through a variety of real-world locations, including houses, restaurants, and universities.

   (b) Develop narratives that incorporate cybersecurity threats, personal challenges, and social pressures, aligning with the game's goals of education and awareness.

2. **Feature 2: Scenario Integration and Milestones**

   (a) Design scenarios that reflect daily life situations, where players face cybersecurity challenges like phishing, social engineering, and other attacks.

   (b) Include milestones and tasks that are contextually relevant to the player's environment, such as securing personal devices at home, protecting data at university, or safely interacting in public spaces.

1.3.5 Module 5: UI/UX

This module focuses on the design of the user interface and the overall user experience, ensuring that the game is intuitive and accessible. It includes the development of storyboards that outline game flow and visual design, enhancing player engagement.

1. **Feature 1: Creation of Game Flow Objects**

   (a) Map out the design of 'mood bar', coins etc.

2. **Feature 2: Design Intuitive Menus and In-Game Prompts**

   (a) Development of clear and start, pause or setting menus, prompts, and feedback systems that guide the player, making navigation and learning straightforward.

1.3.6 Module 6: VR Integration

This module focuses on the virtual reality components of the game, ensuring that all features are optimized for an immersive VR experience. It enhances player engagement by simulating realistic environments and interactions through VR technology.

1. **Feature 1:Create VR-Ready Controls and Interactions**
    (a) Development of controls and interaction methods that are tailored for VR platforms, including support for Oculus Quest 2, allowing players to fully immerse in the game.
2. **Feature 2: Optimizing of 3D Assets and Mechanics for VR**
    (a) Ensures that the game's 3D models, environments, and mechanics are optimized for performance and realism in VR, providing a seamless and engaging player experience.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
Students	Students are the game's primary users. These users will be university students, young people, or even teenagers who must be aware of the cyber security threats they may encounter in virtual environments that mimic real-world locations, including homes, restaurants, and institutions. The age range of players may be 16 to 25. Common cybersecurity problems that students will face include social engineering, USB-based attacks, phishing scams, and more. Decision-making mechanisms and interactive storytelling are all part of the game's strategy to engage this demographic.
Faculty Members	Cybersecurity professionals or experts might use the game for training purposes or to educate young individuals on cybersecurity best practices. Their role is to evaluate the effectiveness of the game in teaching cybersecurity concepts and may provide insights for improvements. They may use the game to simulate real-world cybersecurity incidents for academic or professional training purposes.
Cybersecurity Experts	Experts and professionals in cybersecurity can utilize the game to teach or train youth on cybersecurity best practices. Their role is to evaluate how well the game teaches cybersecurity principles and provide feedback for improvement. For academic or professional training objectives, real-world cybersecurity situations can be replicated in the game. Additionally, the scenarios can be examined by cybersecurity specialists to see if they accurately depict cyberattacks that occur in real life. To make sure that each scenario accurately represents contemporary cybersecurity threats and strategies, they will assess each scenario's authenticity. Experts will also offer suggestions for improvements to guarantee that the game teaches players about appropriate defensive tactics and the consequences of security flaws. Their input will help to improve the scenarios, making them more accurate and consistent with best practices in cyber security awareness.
Game Administrator	Game administrators are responsible for maintaining the game, updating scenarios, and ensuring the VR platform functions properly. They make sure the VR equipment operates properly and that the game is updated with the newest cybersecurity risks. Administrators are in charge of user data, system performance, and game settings.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

Here below are all the story boards and use case diagrams.

2.1.1 Story Board

So first start with story board. Here below is the story board of phishing attack. Where user recieve an email having phishing content . As in start user gets admission in university so it recieved the email having info that register for the Welcome event. So here, The story board is as below here in Fig. 2.1.

[redacted]

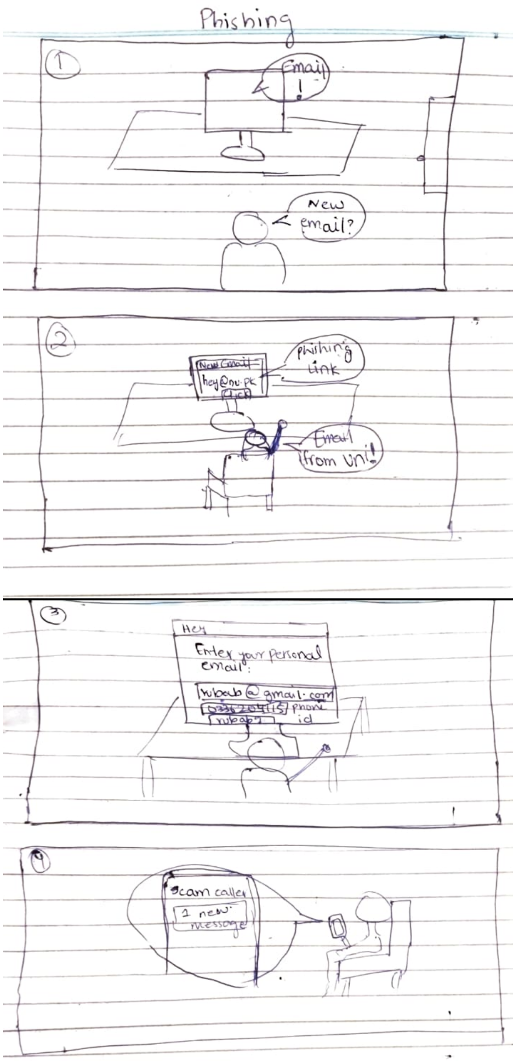

Figure 2.1: Story Board

The Use Case Diagram is as below in Fig. 2.2

2.1.2 Use Case: Start Game

Table 2.1: Start Game

Section	Content
Designation	UC-01-01
Name	Start Game
Authors	Nuzhat Rubab Zahra
Priority	High

Section	Content
Criticality	High
Source	Game Design Team
Responsible	Game designer
Description	The player initiates the game application by launching it on their device, leading them to the main menu where they can start interacting with the game's features.
Trigger Event	The player selects or taps the game icon to launch the application.
Actors	Player
Pre-Condition	• The game is installed and running on the device. • The device is powered on and functioning properly.
Post-Condition	The player enters the main menu of the game and can interact with various options such as "Play Game," "Settings," "View Achievements," etc.
Result	The game environment and main menu are loaded, allowing the player to proceed further.
Main Scenario	1. The player taps the game icon to launch the application. 2. The system initializes the game engine and loads essential game files. 3. The main screen is displayed with the "Start Game" button. 4. The player selects "Start Game." 5. The system loads the main menu, where the player can navigate different options.

Section	Content
Alternative Scenario	• 3a. The game fails to load due to system error or missing files. • 4a. The system prompts the player to reinstall or check for missing files.
Exception Scenario	The game does not start due to external issues such as device power failure or insufficient storage.
Qualities	• QR.01: Load time should not exceed 5 seconds. • QR.02: The transition to the main menu should be smooth and without delays.

2.1.3 Use Case: Main Menu Navigation

Table 2.2: Main Menu Navigation

Section	Content
Designation	UC-01-02
Name	Main Menu Navigation
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	Medium
Source	Game Interface Expert
Responsible	UI/UX Designer
Description	The player interacts with the main menu to navigate different sections such as Profile Menu, Settings, Play Game, etc.
Trigger Event	Player selects a menu option from the main menu.
Actors	Player
Pre-Condition	The game is running, and the main menu is visible.
Post-Condition	Player navigates to the desired section.
Result	Successful navigation to another game section or submenu.

Section	Content
Main Scenario	1. The player is presented with various options in the main menu. 2. Player selects the desired option (e.g., Settings, Profile Menu, Play Game, etc.). 3. System loads the selected submenu. 4. Player is directed to the desired section (Settings, Play Game, etc.).
Alternative Scenario	• 3a. The system encounters a delay in loading the selected section. • 4a. Player is notified and a retry option is provided.
Exception Scenario	The game crashes during the transition to the next screen.
Qualities	• QR.03: Menu responsiveness should be less than 2 seconds. • QR.04: Clear navigation without confusion.

2.1.4 Use Case: Play Game

Table 2.3: Play Game

Section	Content
Designation	UC-01-03
Name	Play Game
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Gameplay Designer
Responsible	Game designer

Section	Content
Description	Player starts gameplay by selecting "Play Game" from the main menu. They are tasked with completing in-game tasks and progressing through the game.
Trigger Event	Player selects “Play Game” from the main menu.
Actors	Player
Pre-Condition	Game is loaded and the main menu is displayed.
Post-Condition	The player enters the gameplay environment.
Result	Player progresses through tasks and interacts with game elements.
Main Scenario	1. Player selects "Play Game" from the main menu. 2. The system loads the current saved game state (or starts a new game). 3. Player is presented with the game environment and objectives. 4. The player starts interacting with the game, completing tasks, and earning rewards.
Alternative Scenario	• 4a. Player fails to complete tasks and restarts from the last checkpoint.
Exception Scenario	The system fails to load the game, causing the game to crash.
Qualities	• QR.05: Smooth and immersive gameplay experience. • QR.06: No freezing or glitches during transitions.

2.1.5 Use Case: Save Game

Table 2.4: Save Game

Section	Content
Designation	UC-01-04

Section	Content
Name	Save Game
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Game State Management Expert
Responsible	Game Designer
Description	The player can pause the game and save their current progress from the Pause Menu.
Trigger Event	The player pauses the game and selects "Save Game" from the pause menu.
Actors	Player
Pre-Condition	The player is actively playing the game and the pause menu is accessible.
Post-Condition	The game is successfully saved, and the player can resume or quit the game.
Result	The player's game progress is saved, and they are returned to the pause menu or the game environment.
Main Scenario	1. The player pauses the game during gameplay. 2. The Pause Menu is displayed. 3. The player selects "Save Game" from the Pause Menu. 4. The system prompts the player to confirm the save. 5. The game state is saved in the player's profile. 6. The player chooses whether to resume the game or quit.
Alternative Scenario	• 4a. The save fails due to insufficient storage space. • 4b. The system informs the player of the failure and requests additional space.
Exception Scenario	The game crashes or the system fails to save due to a hardware error, resulting in loss of unsaved progress.

Section	Content
Qualities	• QR.01: Saving time should not exceed 2 seconds. • QR.02: The player should be able to resume the game seamlessly after saving.

2.1.6 Use Case: Pause Game

Table 2.5: Pause Game

Section	Content
Designation	UC-01-05
Name	Pause Game
Authors	Nuzhat Rubab Zahra
Priority	Medium
Criticality	Medium
Source	Gameplay Designer
Responsible	Game Designer
Description	The player pauses the game to take a break, save the game, access the main menu, or access the game settings.
Trigger Event	Player selects "Pause Game" during gameplay.
Actors	Player
Pre-Condition	Player is in the middle of gameplay.
Post-Condition	Game is paused, and the player can access the pause menu.
Result	The game is paused and the player can choose to resume, save the game, change settings, or quit.
Main Scenario	1. Player presses the pause button. 2. The game freezes the current state and shows the pause menu. 3. The player can choose to resume the game, save the game, adjust settings, or quit to the main menu.

Section	Content
Alternative Scenario	• 3a. Player chooses to go to the main menu.
Exception Scenario	The game crashes when attempting to pause.
Qualities	• QR.09: Smooth transition between paused and resumed states. • QR.10: No data loss during pause.

2.1.7 Use Case: View Extras

Table 2.6: View Extras

Section	Content
Designation	UC-01-06
Name	View Extras
Authors	Nuzhat Rubab Zahra
Priority	Medium
Criticality	Low
Source	Game Design Team
Responsible	Game Designer
Description	The player views the game's extras like achievements, controls, and credits.
Trigger Event	Player selects the extras option from the main menu.
Actors	Player
Pre-Condition	Game is loaded, and the extras menu is accessible.
Post-Condition	Player views the selected extras.
Result	Information is displayed on the screen.

Section	Content
Main Scenario	1. The player navigates to the main menu. 2. The player selects "Extras" from the menu. 3. The system displays options such as "View Achievements," "View Controls," and "View Credits." 4. The player selects the desired option: • 4a. If the player selects "View Achievements," the system shows the list of achievements earned and their descriptions. • 4b. If the player selects "View Controls," the system shows the control layout and button mappings. • 4c. If the player selects "View Credits," the system shows a scrollable list of game credits. 5. The player exits the "Extras" section and returns to the main menu.
Alternative Scenario	• 4a. The player views achievements but hasn't unlocked any, so the system shows a message indicating "No achievements unlocked yet." • 4b. The player views controls and wants to change the control scheme. The system provides an option to modify the control layout directly from the "View Controls" screen. • 4c. The player selects credits, but the credits scroll too fast. The player uses an option to slow down or pause the scrolling.

Section	Content
Exception Scenario	• EX-01: The system fails to load the "Extras" section due to a temporary glitch, causing the game to freeze. The player is prompted to retry or restart the game. • EX-02: The system shows incomplete or corrupted data in the "View Achievements" screen, resulting in a loading error.
Qualities	• QR.11: The extras menu should load within 3 seconds. • QR.12: Achievements, controls, and credits should be presented in a readable format with smooth scrolling or transitions.

2.1.8 Use Case: Manage Profile

Table 2.7: Manage Profile

Section	Content
Designation	UC-01-07
Name	Manage Profile
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	Medium
Source	Game State Management Expert
Responsible	Game Designer
Description	The player manages their profile, including name, and other customization options.
Trigger Event	Player selects "Profile Menu" from the main menu.
Actors	Player
Pre-Condition	The game is running, and the main menu is accessible.
Post-Condition	Player updates profile information successfully.
Result	Profile is updated, and changes are saved.

Section	Content
Main Scenario	1. The player selects "Profile Menu" from the main menu. 2. The system displays the player's current profile information. 3. Player edits profile details (name, character, etc.). 4. The system saves the updated profile.
Alternative Scenario	• 4a. The player cancels changes before saving.
Exception Scenario	• System fails to save the updated profile due to a server error.
Qualities	• QR.13: Profile changes are applied immediately. • QR.14: User-friendly interface for managing profile details.

2.1.9 Use Case: Shop

Table 2.8: Shop

Section	Content
Designation	UC-01-08
Name	Shop
Authors	Nuzhat Rubab Zahra
Priority	Medium
Criticality	Medium
Source	In-Game Purchases Expert
Responsible	Game Designer
Description	The player purchases in-game items or upgrades from the shop.

Section	**Content**
Trigger Event	Player selects "Shop" from the main menu.
Actors	Player
Pre-Condition	The player has sufficient in-game currency.
Post-Condition	The player purchases items, and inventory is updated.
Result	Players successfully purchase items, and they are added to their inventory.
Main Scenario	1. The player selects "Shop" from the menu. 2. The system displays a list of available items for purchase. 3. The player selects an item and confirms the purchase. 4. The system deducts the required amount from the player's currency. 5. The item is added to the player's inventory. 6. 'Purchase' is changed to 'Purchased'.
Alternative Scenario	• 4a. The player cancels the purchase before confirming.
Exception Scenario	• System fails to complete the purchase due to a server error.
Qualities	• QR.15: Purchase confirmation should be instantaneous. • QR.16: User-friendly interface for managing shop items.

2.1.10 Use Case: Earn Rewards

Table 2.9: Earn Rewards

Section	**Content**
Designation	UC-01-09
Name	Earn Rewards
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Reward System Expert
Responsible	Game Designer
Description	The player earns rewards after successfully completing tasks.
Trigger Event	Player completes a task in-game.
Actors	Player
Pre-Condition	Player is actively engaged in completing tasks or objectives.
Post-Condition	Rewards are added to the player's profile or inventory.
Result	Player receives rewards such as points, currency, or unlockables.
Main Scenario	1. The player successfully completes a task or objective. 2. The system calculates the appropriate reward. 3. The reward is added to the player's currency or inventory. 4. The player receives a notification of the earned rewards.
Alternative Scenario	• Player fails to complete the task, and no rewards are given.
Exception Scenario	• System fails to update the player's profile with earned rewards due to a server error.

Section	Content
Qualities	• QR.17: Rewards should reflect immediately in the player's profile. • QR.18: Clear notification of rewards earned.

2.1.11 Use Case: End Game

Table 2.10: End Game

Section	Content
Designation	UC-01-10
Name	End Game
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Game Flow Expert
Responsible	Game Designer
Description	The player finishes their gameplay session and exits the game.
Trigger Event	Player selects "End Game" from the menu.
Actors	Player
Pre-Condition	Player is in the main menu.
Post-Condition	The game exits.
Result	Game session ends.
Main Scenario	1. The player selects "End Game" from the menu. 2. The system prompts the player to confirm they want to exit. 3. The player confirms the action. 4. The game exits to the main menu or desktop.

Section	Content
Alternative Scenario	• Player selects “Cancel” and returns to the game.
Exception Scenario	• The game crashes while closing.
Qualities	• QR.19: Smooth exit without delays.

2.1.12 Use Case: Load Game

Table 2.11: Load Game

Section	Content
Designation	UC-01-11
Name	Load Game
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Game State Management Expert
Responsible	Game Designer
Description	The player loads a previously saved game state.
Trigger Event	Player selects "Load Game" from the main menu.
Actors	Player
Pre-Condition	The player has saved progress in the game.
Post-Condition	The game loads from the saved state, allowing the player to resume from where they left off.
Result	Game state is loaded successfully.
Main Scenario	1. The player selects "Load Game" from the main menu. 2. The system displays a list of saved game states. 3. The player selects the desired save. 4. The system loads the selected game state.

Section	Content
Alternative Scenario	• Player cancels the load process.
Exception Scenario	• System fails to load the saved game state due to data corruption.
Qualities	• QR.20: Loading should be quick and error-free. • QR.21: Ensure integrity of saved data.

2.1.13 Use Case: Resume Game

Table 2.12: Resume Game

Section	Content
Designation	UC-01-12
Name	Resume Game
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	Medium
Source	Game Design Team
Responsible	Game Designer
Description	The player resumes the game after pausing.
Trigger Event	Player selects "Resume Game" from the pause menu.
Actors	Player
Pre-Condition	The game is paused.
Post-Condition	Game resumes from the point it was paused.
Result	The player can continue playing the game.

Section	Content
Main Scenario	1. The player selects "Pause Game" during gameplay. 2. The system freezes game activity and shows the pause menu. 3. The player selects "Resume Game." 4. The system resumes gameplay from the paused point.
Alternative Scenario	• Player chooses to exit to the main menu instead of resuming.
Exception Scenario	• System fails to resume the game due to a freeze or crash.
Qualities	• QR.21: Seamless transition between pause and resume. • QR.22: Quick resume without delays.

2.1.14 Use Case: Defend Against Cyber Attack

Table 2.13: Defend against cyber attack

Section	Content
Designation	UC-01-13
Name	Defend against cyber attack
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Cybersecurity awareness domain expert
Responsible	Game designer

Section	**Content**
Description	The player is attacked by a cyber attacker. They must choose the correct defensive actions based on cybersecurity knowledge to defend themselves.
Trigger Event	A cyber attack is initiated in-game by an NPC attacker.
Actors	Player, NPC Attacker
Pre-Condition	Player is actively progressing in-game.
Post-Condition	Players successfully defend against the attack and maintain their social reputation.
Result	Defence against the cyber attack, leading to point gain.
Main Scenario	1. A cyber attack warning appears on the screen. 2. The player selects an appropriate defensive action (e.g., change password, update software, block attacker). 3. The game evaluates the player's action. 4. The system simulates whether the player successfully defends. 5. If successful, the player gains points and retains reputation. 6. If failed, the player's reputation decreases, and they lose progress.
Alternative Scenario	• 4a. The player ignores the attack. • 4b. The attack causes damage to the player's reputation, and a reset is triggered.
Exception Scenario	• The player has no available tools to defend against the attack, leading to a forced reset.

Qualities	- QR.24: Reaction time upon attack. - QR.25: Difficulty balance of attack.

2.1.15 Use Case: New Game

Table 2.14: New Game

Section	**Content**
Designation	UC-01-014
Name	New Game
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Game Design Team
Responsible	Game Designer
Description	The player starts a new game from the main menu.
Trigger Event	Player selects "New Game" from the main menu.
Actors	Player
Pre-Condition	The game is installed, running, and on the main menu.
Post-Condition	A new game is initiated with default settings.
Result	The game starts, and the player is placed in the first scenario of the first chapter.
Main Scenario	1. The player selects "New Game" from the main menu. 2. The system prompts the player for confirmation. 3. The player confirms to start a new game. 4. The system initializes the game environment and displays the opening sequence.
Alternative Scenario	• 3a. The player cancels and returns to the main menu.

Section	**Content**
Exception Scenario	• The system fails to load the game environment due to a technical issue.
Qualities	• QR.26: Smooth and fast transition to the game environment. • QR.27: No delays longer than 5 seconds when starting a new game.

2.1.16 Use Case: Continue Game

Table 2.15: Continue Game

Section	**Content**
Designation	UC-01-015
Name	Continue Game
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Game Design Team
Responsible	Game Designer
Description	The player continues from the currently left game state.
Trigger Event	Player selects "Continue Game" from the main menu.
Actors	Player
Pre-Condition	There is a saved game state.
Post-Condition	The player resumes the game from the last saved state.
Result	The currently left game is loaded, and the player continues from their previous progress.

Section	Content
Main Scenario	1. The player selects "Continue Game" from the main menu. 2. The system loads the currently left game state. 3. The player resumes the game from where they left off.
Alternative Scenario	• 2a. The system cannot find the currently left game state and prompts the player to start a new game.
Exception Scenario	• The system fails to load the currently left game state due to corruption or missing files.
Qualities	• QR.03: Fast loading of the currently left state, within 5 seconds. • QR.04: Ensure no data loss during the loading process.

2.1.17 Use Case: Select Chapter

Table 2.16: Select Chapter

Section	Content
Designation	UC-01-16
Name	Select Chapter
Authors	Nuzhat Rubab Zahra
Priority	Medium
Criticality	Medium
Source	Game Design Team
Responsible	Game Designer
Description	The player selects a specific chapter to play from.
Trigger Event	Player selects "Select Chapter" from the main menu.
Actors	Player

Section	Content
Pre-Condition	Chapters have been unlocked through previous gameplay.
Post-Condition	The selected chapter is loaded.
Result	The player starts the selected chapter.
Main Scenario	1. The player selects "Chapters" from the main menu. 2. The system displays the available chapters. 3. The player replays or continues a chapter. 4. The system loads the chosen chapter.
Alternative Scenario	• 3a. The player cancels the selection and returns to the main menu.
Exception Scenario	• The system fails to load the selected chapter due to missing files.
Qualities	• QR.28: Chapters load within 3-5 seconds. • QR.29: Ensure only unlocked chapters are selectable.

2.1.18 Use Case: Complete Task In Story

Table 2.17: Complete Tasks in Chapter

Section	Content
Designation	UC-01-17
Name	Complete Tasks in Chapter
Authors	Nuzhat Rubab Zahra
Priority	High
Criticality	High
Source	Game Design Team
Responsible	Game Designer

Section	Content
Description	The player completes a series of tasks or objectives within a chapter of the game.
Trigger Event	The player advances to a new chapter and is presented with tasks to complete.
Actors	Player
Pre-Condition	The chapter has started, and the task objectives are available to the player.
Post-Condition	The player successfully completes the tasks, and the chapter progress is updated.
Result	Tasks are completed, and progress in the current chapter is saved.
Main Scenario	1. The player starts a new chapter or resumes a previously saved chapter. 2. The system presents the list of tasks or objectives for that chapter. 3. The player completes the first task. 4. The system confirms the task completion and updates the task list. 5. The player proceeds with subsequent tasks, repeating steps 3 and 4. 6. Upon completing all tasks, the system updates the chapter progress. 7. Player is rewarded for successfully completing the chapter tasks.
Alternative Scenario	• 4a. The player skips or fails a task, causing a delay in chapter progression. The player is given the option to retry the task or skip it.

Section	Content
Exception Scenario	• The system fails to update the task completion due to a bug or server issue, causing progress to halt. The game offers the option to restart the chapter or reload the last saved checkpoint.
Qualities	• QR.30: Task progress is automatically saved in real-time to avoid loss of progress. • QR.31: Clear and intuitive user interface for displaying and tracking task objectives. • QR.32: No performance drop during task transitions or task completion events.

Storyboarding for graphically intensive applications.
Create the suitable representation here.

2.2 Functional Requirements

This section outlines the functional requirements of the cybersecurity awareness game, focusing on features, player interactions, and game mechanics.

2.2.1 Module 1: User Interface (UI) & User Experience (UX)

The following requirements focus on the visual and interactive aspects of the game:

1. **FR1**
   The system shall provide a simple, intuitive, and consistent user interface for navigating menus, settings, and indirect story-based tutorials.
2. **FR2**
   The game shall offer real-time feedback in response to player actions, displayed visually on the screen.
3. **FR3**
   The game shall allow the player to view the basic information about his overall

performance, level, and scores.

4. **FR4**
   The system shall include a "Mood Bar" that dynamically reflects the player's social status using two states: "happy" and "dead inside."

5. **FR5**
   The system shall guide players through challenges by providing contextual prompts and suggestions.

6. **FR6**
   The audio subsystem shall adjust ambient sounds and music based on player interactions and environments to enhance immersion.

2.2.2 Module 2: Virtual Reality Integration

This section describes requirements for integrating VR systems:

1. **FR1**
   The system shall be compatible with VR headsets, including Oculus, allowing full VR gameplay.

2. **FR2**
   VR environments shall simulate real-world locations like buses, universities, and restaurants, allowing players to interact with them.

3. **FR3**
   The system shall allow players to interact with non-player characters (NPCs) in the VR environment through dialogue and decision-making interfaces.

4. **FR4**
   Player decisions in virtual environments shall dynamically change the scenario and storyline.

2.2.3 Module 3: Cybersecurity Attack Simulation

The game simulates cyberattacks and teaches players how to respond to them:

1. **FR1**
   The game shall simulate cyberattacks such as phishing, MitM, brute force, malware, and ransomware, presented within real-life daily scenarios.

2. **FR2**
   The system shall provide educational content for each simulated cyberattack, explaining its nature and preventive measures.

3. **FR3**
   After a successful cyberattack, the system shall provide players with damage control strategies to recover from the incident.

4. **FR4**
   The system shall restrict certain cyberattack simulations, such as DDoS or SQL injections, which are not within the educational scope of the game..

2.2.4 Module 4: Reputation & Social Points System

The player's reputation and social points change based on in-game decisions:

1. **FR1**
   The system shall adjust the player's reputation based on their response to social interactions and cyberattacks.

2. **FR2**
   The game shall reward positive actions with social points and in-game currency (coins) and penalise negative actions by reducing social points.

3. **FR3**
   The "Mood Bar" shall reflect the player's accumulated social points and visually indicate their emotional state.

4. **FR4**
   Losing all social points shall result in failure of the current level, prompting a restart from a checkpoint.

2.2.5 Module 5: Educational Content & Tutorials

The game includes educational content to teach players about cybersecurity threats:

1. **FR1**
   The game shall provide real-time feedback and hints to help players learn and reinforce proper responses to cyberattacks.

2. **FR2**
   After an attack, the system shall give preventive tips to avoid similar attacks in future levels.

2.2.6 Module 6: Game Progression & Levels

This section describes how the player advances through the game:

1. **FR1**
   The game shall consist of multiple levels, with each level representing a different phase of the player's life, such as university or early career.
2. **FR2**
   Players shall unlock new virtual environments (homes, offices, public places) by completing tasks and reaching milestones.
3. **FR3**
   Each level shall present increasingly complex challenges requiring players to apply their knowledge of cybersecurity.

2.2.7 Module 7: Coins & Rewards System

Players are rewarded for good decisions with in-game currency:

1. **FR1**
   The game shall reward players with coins for making good decisions, such as protecting personal data or successfully avoiding a cyberattack.
2. **FR2**
   Players shall use earned coins to unlock in-game items or progress to higher levels.
3. **FR3**
   The system shall provide visual and auditory feedback when coins are earned to reinforce positive behaviour.

2.2.8 Module 8: Decision-Making & Scenario Selection

Player decisions affect the game's storyline and outcomes:

1. **FR1**
   The system shall present players with decisions that impact the game's storyline, affecting their social standing, relationships, and future challenges.
2. **FR2**
   The outcome of these decisions shall dynamically alter the next set of challenges, creating a unique experience based on the player's choices.

3. **FR3**
   The player's decision-making process shall influence the behaviour of NPCs, affecting their responses and interactions with the player.

2.2.9 Module 9: Non-Player Characters (NPCs) Interaction

This section outlines how players interact with NPCs:

1. **FR1**
   NPCs in the game shall exhibit different behaviours based on the player's decisions, reflecting real-world social dynamics (e.g., rivals, friends, and family).
2. **FR2**
   NPCs shall act as attackers, mentors, or allies depending on the context and scenarios within the game.
3. **FR3**
   The player shall interact with NPCs through dialogues, activities, and decision points that affect their social standing.

2.2.10 Module 10: Task & Milestone Management

The following requirements address task-based gameplay:

1. **FR1**
   The game shall feature milestones like reaching academic or personal goals (e.g., completing tasks in university life or office life).
2. **FR2**
   The system shall present challenges and tasks in each level that must be completed to move forward in the game, simulating real-life responsibilities and challenges.
3. **FR3**
   Each task must integrate a learning component related to cybersecurity, such as completing a challenge without falling for a phishing scam or maintaining privacy during online interactions.

2.2.11 Module 11: Level Failure & Restart Mechanism

The following are the requirements for game progression and failure:

1. **FR1**
   Players shall lose a level if they reach the "dead inside" state on the mood bar, after which they must restart the level.

2. **FR2**
   The game shall provide checkpoints within levels so that players don't lose complete progress upon failure.

3. **FR3**
   The system shall offer tips or guidance when players fail a level, helping them understand what went wrong and how to avoid failure in the next attempt

2.2.12 Module 12: Daily Life Simulation

The following requirements enhance the realistic portrayal of daily life:

1. **FR1**
   The game shall simulate various daily activities, such as travelling on a bus, attending classes, or visiting restaurants.

2. **FR2**
   These daily activities must be interwoven with cybersecurity education, where the player faces potential risks (e.g., using free Wi-Fi at a café or sharing information with strangers).

3. **FR3**
   The player's interaction with the virtual world (e.g., environments like homes or restaurants) shall mirror real-life cybersecurity challenges and solut ions.

2.2.13 Module 13: Level Save Progression Management

The following requirements manage player progress and saving levels:

1. **FR1**
   The game shall automatically save player progress at predefined checkpoints, such as the completion of tasks, levels, or significant decisions within the storyline.

2. **FR2**
   Players must have the option to manually save their current progress at any point in the game through an easily accessible "Save" button in the user interface.

3. **FR3**
   The game shall allow players to load their saved progress from the main menu or during the game.

4. **FR4**
   The game shall include a checkpoint system that informs players when their progress is automatically saved.

5. **FR5**
   If a player attempts to save during a critical event (e.g., during a cyberattack or key decision-making moment), the system shall prompt the player to complete the event before allowing the save.

2.2.14 Module 14: Performance

1. **FR1**
   The player shall be able to view their current performance.

2. **FR2**
   The player shall be able to interact with the updated gaming based on their performance.

2.2.15 Module 15: Ending

1. **FR1**
   The game shall end once the player successfully reaches their office milestone.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.3.1 Reliability

REL-1: The game shall recover from unexpected crashes or disruptions within 10 seconds and restore the player's progress to the last saved checkpoint.

2.3.2 Usability

Usability requirements deal with ease of learning, ease of use, error avoidance and recovery, the efficiency of interactions, and accessibility. The usability requirements specified here will help the user interface designer create the optimum user experience.

USE-1: The VR system shall allow users to easily navigate the game menus using simple hand gestures or VR controllers.

USE-2: The game must be simple and have a user-friendly interactive interface.

USE-3: The game shall provide on-screen prompts to guide players through learning objectives and gameplay without external assistance.

USE-4: The game must display error messages related to gameplay e.g., invalid actions.

USE-5: The game must allow users to pause, save, and resume their progress at any time to accommodate players with varying time constraints or play preferences.

2.3.3 Performance

PER-1: The game must load scenarios within 10 seconds.

PER-2: System shall perform well for continuously evaluating the performance of the user.

PER-3: System shall be responsive towards the user's actions

PER-4: The system shall ensure that all in-game data processing, such as updating scores or generating cybersecurity attack results, is completed within **1 second** to maintain a real-time experience.

2.3.4 Security

SEC-1: Profile of one user shall not be accessed by any other user

2.4 Domain Model

The Domain Model is displayed as Fig. 2.3 below:

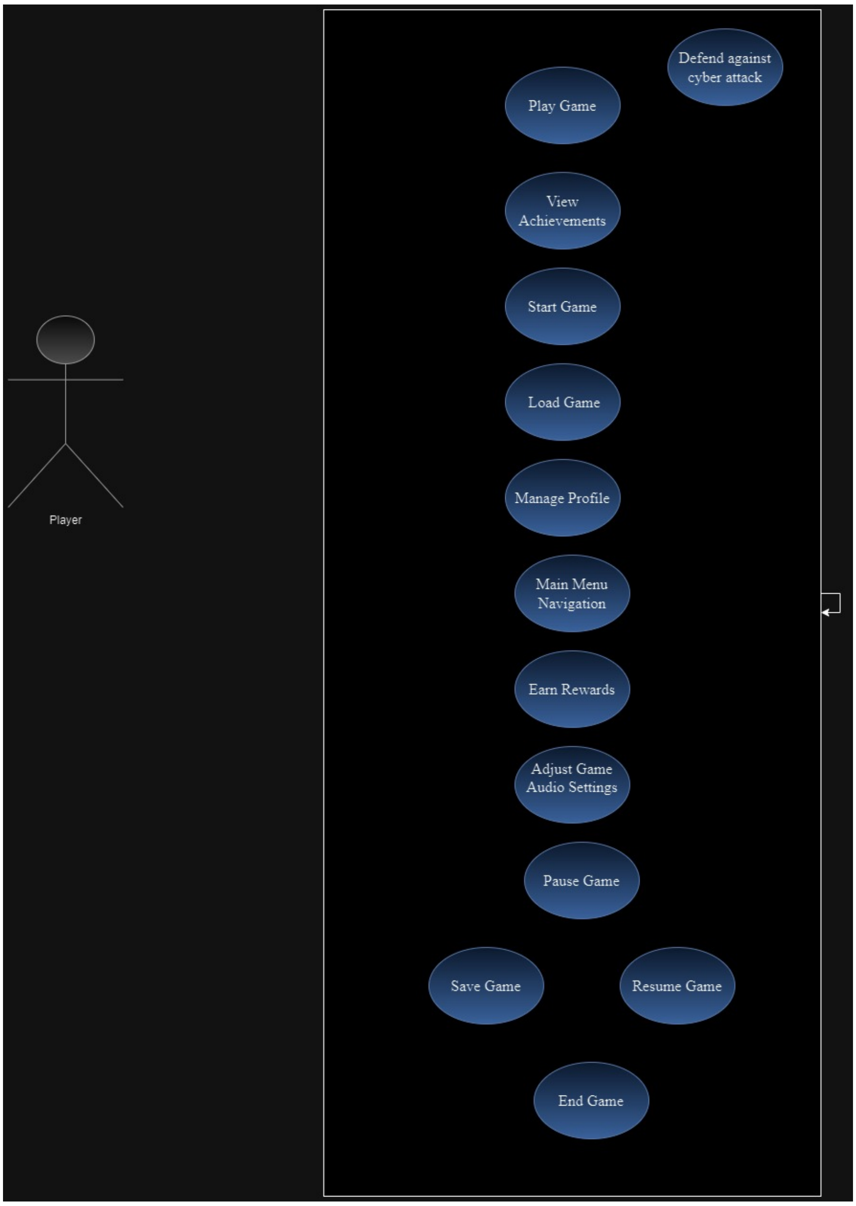

Figure 2.2: Use Case Diagram

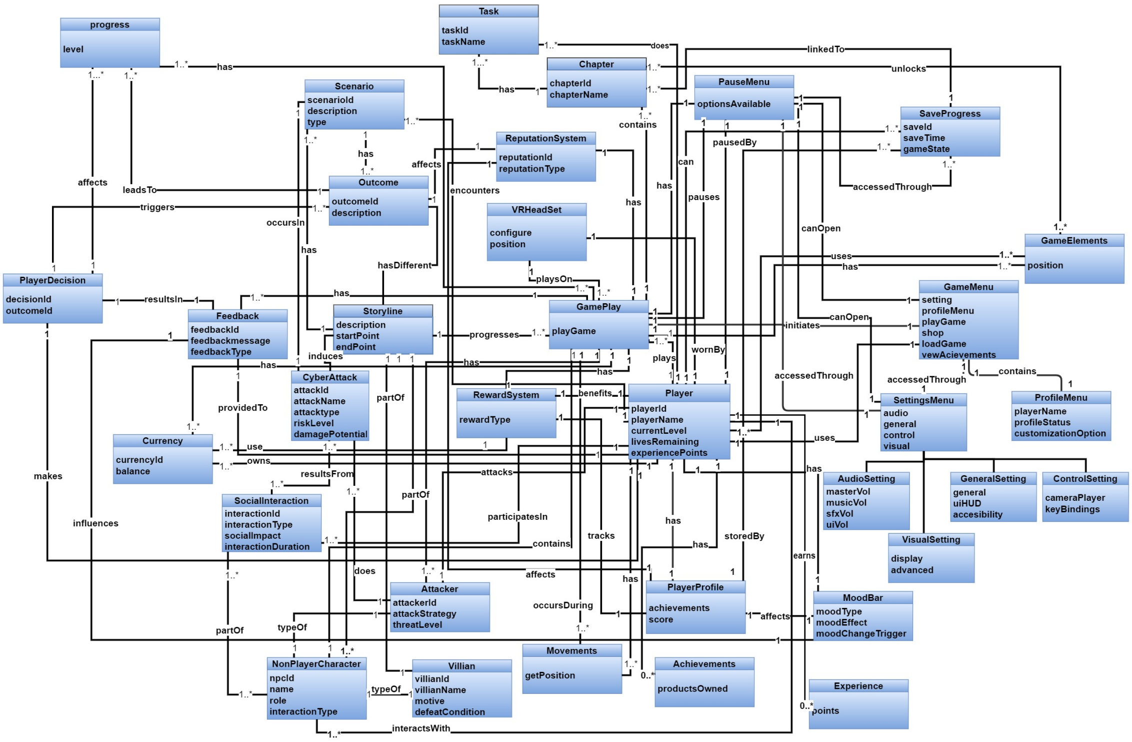

Figure 2.3: Domain Model

Chapter 3

System Overview

The Cybersecurity Awareness Game is an innovative educational game aimed at teenagers and young adults, designed to raise awareness about cybersecurity threats and promote safe online practices. The game harnesses the power of virtual reality (VR) technology to create an immersive experience that mimics real-life situations, allowing players to engage with various scenarios they might encounter in their everyday lives—such as interacting on social media, attending classes, or navigating online communications.

Functionally, the game integrates a user-friendly interface that guides players through different levels while providing real-time feedback based on their actions. Players navigate through challenges that simulate various cybersecurity threats, such as phishing attacks, malware, and social engineering, learning preventive measures and appropriate responses in the process. A dynamic reputation system tracks the player's decisions, affecting their in-game reputation and social standing, thereby reinforcing the impact of their choices on their digital lives.

Contextually, the game is developed to resonate with daily experiences of its target audience, making the learning experience relatable and engaging. The design prioritises both fun and functionality, ensuring that players are not only entertained but also educated about essential cybersecurity principles in a safe, interactive environment. This project seeks to address the growing need for effective cybersecurity education among young users, equipping them with the knowledge and skills to navigate the digital world confidently and securely.

3.1 Architectural Design

The Architectural design is displayed in Fig. 3.1 below:

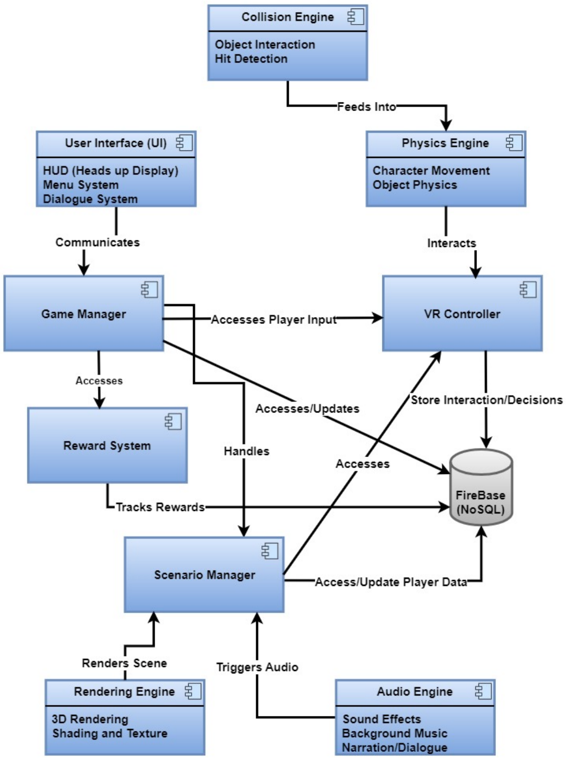

Figure 3.1: Architectural Design

3.2 Design Models

Design Models for Procedural Approach

The applicable models for the project using procedural approach may include:

3.2.1 Activity Diagram

The Activity diagram is displayed in Fig. 3.2 below:

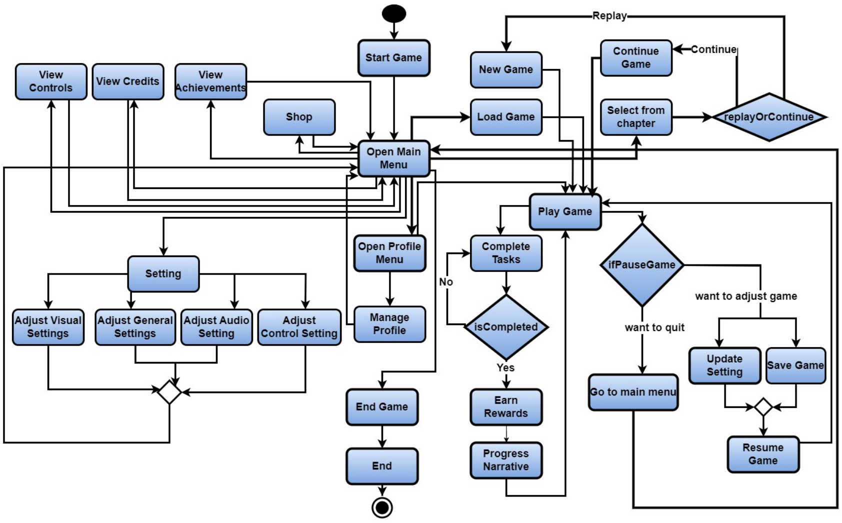

Figure 3.2: Activity Diagram

3.2.2 System-level Sequence Diagram

The System-level Sequence Diagram is displayed as Fig. 3.3 below:

3.2.3 Data Flow Diagram

3.2.3.1 Level 0

1. **Player** The player is an external entity that interacts with the system by inputting choices or decisions related to cyber security scenarios (e.g., phishing attack, USB-based attack).

2. CyberQuest (central process) The game receives inputs from the player, processes those inputs, and sends data back to the player as feedback or scores.

3. **Firebase** The database system that stores player data, such as progress, scores, decisions, and choices. Firebase interacts with the game to store and retrieve information.

4. **VR System** The VR component provides the immersive experience and retrieves player actions to send them back to the game system for decision-making.

3.2.3.2 Level 0 DFD

1. **Player -> CyberQuest** The player sends actions and receives feedback.

2. **CyberQuest -> Firebase** The game reads/writes player data (progress, scores, etc.).
3. **Game -> VR System** The game receives VR input (e.g., actions performed in VR scenarios) and sends updates back to the VR environment.

3.2.3.3 Level 1 DFD (Expanding Processes)

1. **Decision-Making Process**

   Players input decisions related to different scenarios. Decisions are stored in Firebase, and outcomes are calculated.
2. **Game Feedback Process** Based on decisions made by the player, feedback (score, advice) is generated. The Firebase is updated with new player data (scores).
3. **VR Interaction Process** Player actions in VR (e.g., navigating a cybersecurity scenario) are processed and fed into the decision-making process.
4. **Data Management (Firebase)** This handles storing and retrieving all game data (progress, actions, player profiles).

3.2.3.4 Level 1 DFD

1. **Player** -> **Decision-Making Process** The player makes choices that affect the game.
2. **Decision-Making Process** -> **Game Feedback Process** The game evaluates the player's decisions.
3. **VR Interaction Process** -> **Decision-Making Process** VR interactions (player actions in the VR world) are fed into the game for evaluation.

The Data Flow Diagram is displayed as Fig. 3.4 below:

3.2.4 State Transition Diagram

Steps:

1. **Initial State** The player begins unaware of potential cyber-attacks.
2. **States**
   (a) Daily Routine: The player goes through daily activities such as checking email, visiting the university, or going out to enjoy.

(b) Scenario Encounter: The player faces a situation where they are potentially exposed to a cyber threat (e.g., phishing, social engineering).

(c) Decision Point: The player must make a choice, a critical moment where they can avoid or fall victim to the attack.

(d) Attack Occurs: If the wrong decision is made, a cyber attack happens (e.g., phishing or malware attack).

(e) Safe: If the right decision is made, the player avoids the attack.

(f) Feedback & Learning: The player receives feedback about the result and learns about the consequences of their choices.

(g) Continue Game: After feedback, the player continues or ends the game and is now more aware of potential threats.

3. **Transitions**

(a) From the Daily Routine to Scenario Encounter.

(b) From Scenario Encounter to Decision Point.

(c) From Decision Point to either Attack Occurs or Safe.

(d) From Attack Occurs to Feedback & Learning.

(e) From Feedback & Learning to Continue Game or Game Over.

The State Transition Diagram is displayed as Fig. 3.5 below:

3.3 Data Design

The "Game-Based Intervention for Cyber Security Awareness" system is organised, stored, and processed. The system uses Firebase, a NoSQL database, to manage player data, scenarios, decisions, and other game-related entities.

3.3.1 Data Storage Solution

The system uses Firebase (NoSQL) for data storage. Firebase is perfect for handling real-time data synchronisation, managing complex relationships, and providing scalability, which is crucial for a VR-based game. Additionally, Firebase supports user authentication, making it easier to manage player profiles and ensure secure access.

3.3.2 Data Entities and Structure

The major data entities in the system include **UserProfile**, **Scenario**, UserDemographics, and **DecisionLog**. Each entity is stored in Firebase in a structured and scalable manner. Below is a breakdown of each entity and its corresponding fields:

1. **UserDemographics**
   At the start of the game, the player will provide their **username** and **gender**. This entity stores basic information about the player.

   - **userID** (string): Unique identifier for each player (auto-generated by Firebase).
   - **username** (string): The player's username or player name.
   - **gender** (string): The player's gender (e.g., "Male," "Female," "Other"). This data helps personalise the player's experience and can be used to tailor game content or display gender-specific avatars and preferences.

2. **UserProfile**
   This entity stores player-specific information, including their progress, score, reputation in the game and profile data.

   - **userID** (string): Unique identifier for each player (UserDemographics).
   - **username** (string): The player's username or player name.
   - **cyberSecurityScore** (int): Player's cumulative score based on their decisions and progress in the game.
   - **progress** (object): Tracks the current level, completed scenarios, and reputation status.
     - **currentLevel** (int): The level the player is currently on.
     - **reputationScore** (int): Player's reputation score affected by their decisions in-game.
     - **mood** (string): Current mood (e.g., "happy" or "dead inside").
     - **completedScenarios** (array of strings): List of scenario IDs that the player has completed.
   - **achievements** (array of objects): Stores special achievements unlocked by the player.

3. **Scenario**
   This entity defines the various scenarios in the game, including the type of cyber attack involved and the possible outcomes based on player decisions.

   - **scenarioID** (string): Unique identifier for each scenario.

- **scenarioTitle** (string): The title or name of the scenario (e.g., "Phishing Attack at University").
- **description** (string): Detailed description of the scenario.
- **threatType** (string): Type of cyber threat demonstrated in the scenario (e.g., Phishing, Malware).
- **possibleOutcomes** (array of objects): Different possible results based on player decisions.
  - **outcomeID** (string): Unique ID for each outcome.
  - **decisionRequired** (string): The decision that the player must make (e.g., "Open suspicious email").
  - **result** (string): The result of the decision (e.g., "Player's personal data gets leaked").
  - **reputationImpact** (int): The impact of the decision on the player's reputation score.
  - **moodImpact** (int): How the decision affects the player's mood (positive or negative).
  - **cyberSecurityKnowledge** (string): Educational information provided after the outcome is reached.

4. **DecisionLog**
   The DecisionLog stores records of the player's decisions and the corresponding outcomes. This information helps track the player's learning journey and game progression.

   - **logID** (string): Unique identifier for the decision log entry.
   - **userID** (string): ID of the player who made the decision (links to UserProfile).
   - **scenarioID** (string): ID of the scenario being played (links to Scenario).
   - **decision** (string): The decision made by the player.
   - **outcome** (string): The resulting outcome of the decision.
   - **timeStamp** (datetime): The timestamp when the decision was made.

3.3.3 Database Structure in Firebase

The Firebase database is organised into collections and documents. Each entity is represented as a collection of documents, making it easy to query and store data for each player, scenario, and decision. Example Firebase structure:

```
/UserDemographics
   /userID1
      username: "JohnDoe"
      gender: "Male"

/UserProfile
   /userID1
      cyberSecurityScore: 85
      progress: {
          currentLevel: 3,
          reputationScore: 65,
          mood: "happy",
          completedScenarios: ["scenarioID1", "scenarioID2"]
      }
      achievements: [
          {name: "Phishing Survivor", date: "2024-10-02"},
          {name: "Malware Preventer", date: "2024-10-05"}
      ]

/Scenario
   /scenarioID1
      scenarioTitle: "Phishing Attack"
      description: "Player receives a phishing email..."
      threatType: "Phishing"
      possibleOutcomes: [
         {outcomeID: "outcomeID1", decisionRequired: "Click link", result: "Player
         {outcomeID: "outcomeID2", decisionRequired: "Ignore email", result: "Playe
      ]

/DecisionLog
   /logID1
      userID: "userID1"
      scenarioID: "scenarioID1"
      decision: "Click link"
      outcome: "Player data gets stolen"
      timeStamp: "2024-10-02T10:15:00Z"
```

3.3.4 Data Processing and Organization

- **Firebase Collections**: Each entity is represented as a collection in Firebase (e.g., UserDemographics, UserProfile, Scenario, DecisionLog). Firebase's NoSQL structure allows for flexible, dynamic data storage and efficient querying.

- **Player Onboarding:** The UserDemographics collection stores the username and gender provided by players during the onboarding process. This data is linked to the player's UserProfile, tracking their progress, score, and reputation throughout the game.

- **Player Progress**: Each player's progress, including completed scenarios, decisions, and learning outcomes, is logged in their profile in the UserProfile collection, providing a personalised gaming experience.

- **Scenario Tracking**: Each scenario is stored in the scenario collection, and the outcomes are dynamically loaded based on the player's decisions.

- **Real-time Updates**: Firebase ensures real-time syncing for continuous player progress updates and immediate logging of scenario outcomes.

- **Decision Logs**: The DecisionLog collection records each decision made by the player, enabling tracking of their learning journey and allowing for future analysis.

- **Scenario Execution**: When a player interacts with a scenario, the system loads data from the scenario collection. After the player makes a decision, the corresponding outcome is stored in the DecisionLog, and the player's UserProfile is updated with the new progress and reputation score.

  This data design is optimised to ensure a smooth and personalised experience for each player in the VR game, with a focus on scalability, performance, and real-time interaction.

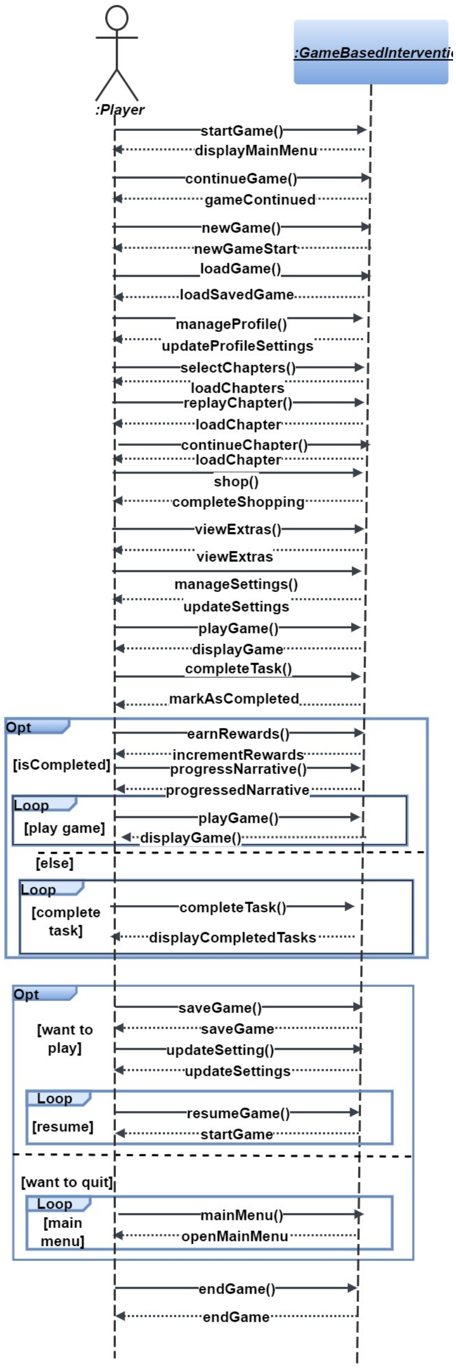

Figure 3.3: SSD

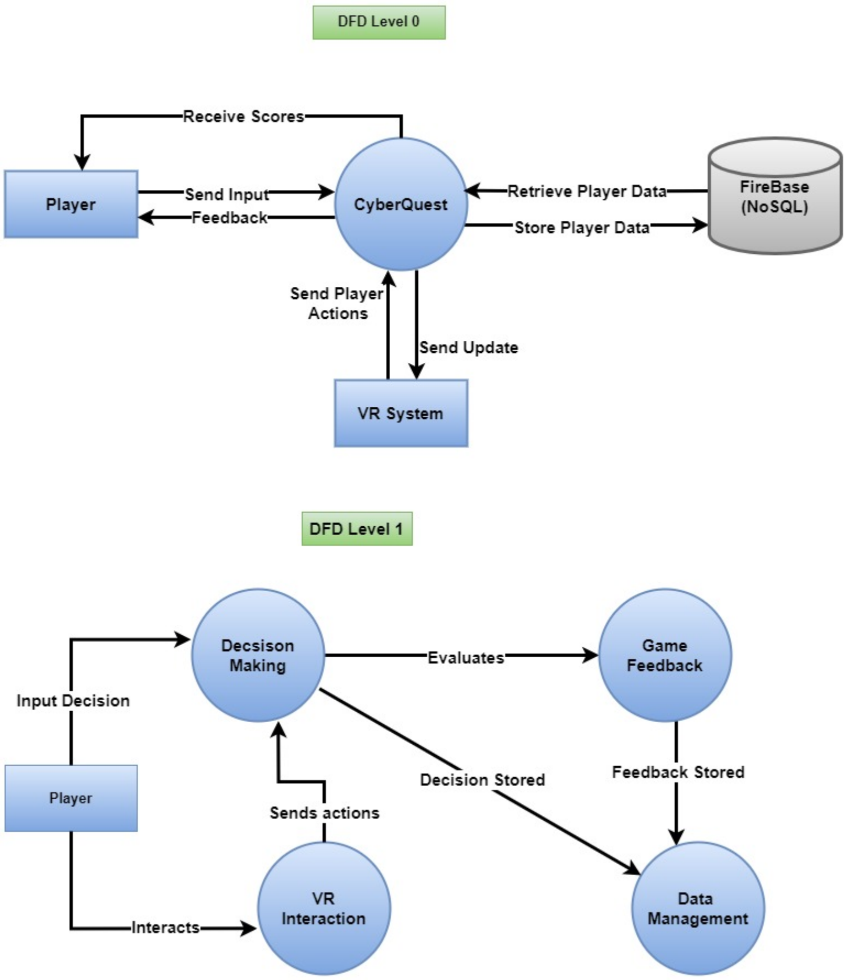

Figure 3.4: Data Flow Diagram

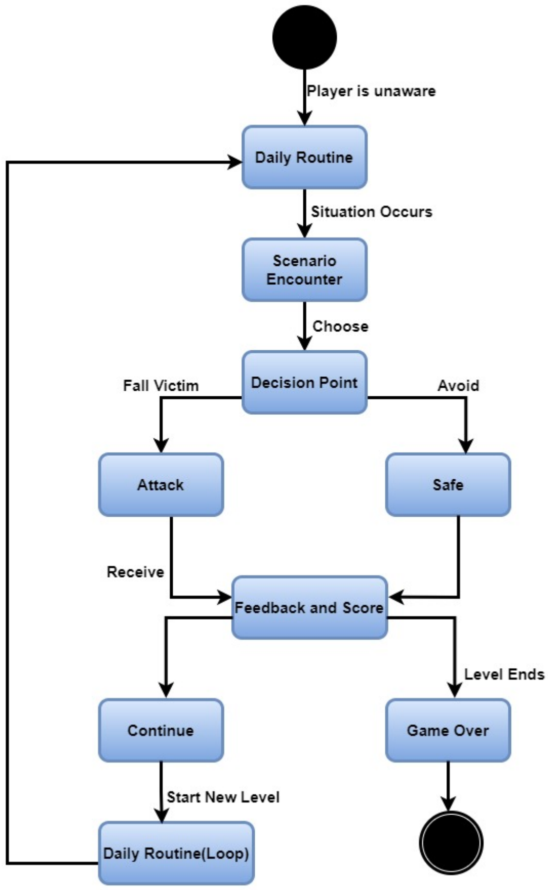

Figure 3.5: State Transition Diagram